2. Data Specification

2.1 Data Used

This paper will use the following data as representations of the U.S equity and fixed-income markets:

- Equity Market: SPDR S&P 500 ETF Trust (SPY) - daily adjusted closing prices. It is chosen because of its high liquidity and direct replication of the S&P 500 index.
- Fixed-Income Market: ishares 20+ Year Treasury Bond ETF (TLT) - daily adjusted closing prices. TLT is selected because it exhibits the highest sensitivity to the Federal Reserve rate decisions making it the most directly affected bond during the 2022 rate-hiking cycle

The use of Exchange-Traded funds or ETFs represent tradable instruments that reflect real portfolio exposure to the stock-bond correlation consistent with the 60/40 portfolio framing.

2.2 Data Frequency and Source

The Daily adjusted closing prices will be sourced from Yahoo Finance using the yfinance python library.

2.3 Return Computation

Both price series will be transformed into log-returns prior to any analysis which is computed as

$$R_t = \ln(P_t / P_{\{t - 1\}})$$

Log-returns are selected over simple percentage returns because they are the standard-convention in the empirical finance literature (Campbell, Lo & MacKinlay, 1997).

2.4 Sample Period

The full sample covers January 2, 2021 to December 30, 2024 subdivided as follows:

Pre-Shift Baseline	Jan 2021 – Dec 2021	Negative correlation; Gaussian-like returns
Regime-Shift Period	Jan 2022 – Jul 2022	Active correlation reversal; peak rate-hike pace
Sustained High-Rate Regime	Aug 2022 – Dec 2023	Positive correlation stable; rate hikes decelerating then pausing

In the pre-shift baseline, it suggests a calm period where no crashes happened. This creates a Gaussian-like distribution which we will use to compare to the signals in the noisy market crash during the Fed Rate hikes

3. Research Questions and Formal Null Hypotheses

Each research question is accompanied by a testable null hypothesis (H_o) and alternative hypothesis (H_1). Rejection of H_O suggests empirical support for the thesis argument

RQ1: Signal Stability (Statistics)

Research Question: To what extent does Effective Transfer Entropy (ETE) provide a more stable causal signal, measured by a lower coefficient of variation (CV) and divergence behaviour in rolling window estimates compared to Linear Granger Causality (LGC) during the 2022 stock-bond regime shift?

Coefficient of Variation will be solved by dividing the standard deviation with the mean which represents the variability relative to the mean. A lower CV indicates better stability. This is used to tackle the difference between the units that both ETE and LGC output.

H_o: The Coefficient of Variation(CV_ETE) of rolling-window ETE estimates is not significantly lower than the Coefficient of Variation of rolling window LGC Estimates (CV_LGC) during the 2022 regime shift period

H_1: CV_ETE < CV_LGC during the 2022 regime shift period, indicating that ETE provides a more stable causal signal than LGC under structural change

RQ2: Non-Linearity Identification (Statistics)

Research Question: Does the divergence between normalized ETE and LGC estimates increase significantly during the 2022 regime shift relative to the 2021 baseline, indicating the emergence of higher-order, non-linear risk transmission dynamics

Divergence will be solved by getting the absolute difference of the normalized scores of ETE and LGC per rolling window

H_0: Mean absolute normalized divergence between ETE and LGC rolling estimates is not significantly greater in 2022 than 2021

H_1: The mean absolute normalized divergence between ETE and LGC is significantly greater during the 2022 regime shift period than during the 2021 baseline, indicating that non linear information flows are present and that LGC fails to capture them

RQ3: Algorithm Implementation and Benchmarking (Computer Science)

Research Question: Can an Effective Transfer Entropy based regime detection improve risk control and drawdown mitigation relative to a 60/40 stock-bond allocation and LGC Regime Detection Algorithm during the 2022 stock-bond structural break?

H_o: ETE-adaptive allocation does not significantly reduce realized volatility, maximum drawdown or sharpe ratio relative to the 60/40 stock-bond allocation model during 2022

H_1: The ETE-adaptive allocation model produces lower realized volatility and/or reduced maximum drawdown and sharpe ratio during the 2022 regime shift

RQ4: Algorithm Benchmarking (Computer Science)

Research Question: Does the Effective Transfer Entropy Regime Detection Algorithm detect the regime shift earlier than Linear Granger Causality Regime Detection Algorithm.
H_o: The ETE-RDA regime detection date is not significantly earlier than the LGC-RDA detection date
H_1: The ETE-RDA detects the regime shift at least n trading days earlier than LGC-RDA, where n is determined by the rolling Pearson correlation crossing zero as the reference anchor.

RQ5: Parameter Sensitivity (Computer Science)

Research Question: How does the choice of time lag (k) and embedding dimension impact the stability of ETE and LGC estimates when applied to non-stationary financial time series?

H_0: The stability metrics (CV, Kurtosis, Directional Consistency) of ETE and LGC do not significantly change across lag values $k \in \{1, 2, 3, 5\}$ and window sizes $n \in \{20, 40, 60\}$.

H_1: The stability metrics of ETE are less sensitive to changes in k and n than those of LGC, indicating that the non-parametric ETE estimator is more robust to parameter choice under non-stationary conditions.

4. Pre-Processing and Diagnostics

Before we estimate and compute for ETE and LGC, we must run a diagnostic test on the data to ensure it fits the conditions we need to run the tests

4.1 Stationarity Testing

To check if all return series are stationary, we will be employing two complementary tests.

1. Augmented Dickey-Fuller (ADF) Test: checks the presence of a unit root. The presence of a unit root indicates non-stationarity
2. Kwiatkowski-Philips-Schmidt-Shin (KPSS) Test: Checks for stationarity around a deterministic trend. This test is used to complement the ADF test

Stationarity will be verified for each rolling window through the ADF test. Windows failing the stationarity criterion will be flagged, and their LGC estimates will be excluded from the stability analysis to prevent LGC estimates from being contaminated, ensuring fair comparisson.

4.2 Cointegration TEsting

If both SPY and TLT log-return series are individually found to be non-stationary (I(1)), a Johansen cointegration test will be conducted to determine whether a long-run equilibrium

relationship exists between the two series. If cointegration is confirmed, the appropriate model for LGC estimation is the Vector Error Correction Model (VECM) rather than the standard VAR. This distinction is critical: using a VAR on cointegrated series omits the error-correction term, producing a misspecified model and biased Granger causality estimates. The thesis will document this decision with the cointegration test result.

4.3 Normality Testing

This step is the empirical foundation for the thesis's central claim. The Barnett-Barrett-Seth (2009) theorem establishes that LGC ≡ ETE under Gaussianity. Therefore, demonstrating that the 2022 return distributions deviate from Gaussianity is not merely contextual **it is the mechanism that explains why the two measures diverge.**

A rolling Jarque-Bera (JB) test will be applied to each rolling window of returns for both SPY and TLT, producing a time series of p-values for the normality hypothesis. The evolution of JB p-values from 2021 through 2022 will be plotted to show the progression from Gaussian to non-Gaussian return conditions, visually and statistically corroborating the regime shift.

Additionally, excess kurtosis of the full 2022 return distributions will be reported, as fat tails are the primary source of non-Gaussianity in financial returns (Cont, 2001).

5. Estimation Procedures

5.1 Transfer Entropy Estimation

Transfer entropy (TE) is measured to quantify directional information flow between time series using methods like histogram-based estimation, K-Nearest Neighbors (KNN), kernel density estimation, and adaptive partitioning.

In this paper our ETE Estimation will utilize the Kraskov-Stögbauer-Grassberger (KSG) k-nearest-neighbor entropy estimator. KSG is the standard Transfer Entropy Estimator which takes inspiration from the Kozachenko-Leonenko Entropy Estimator.

5.2 ETE: Noise Floor Via Surrogate Data

To implement Effective Transfer Entropy from its raw TE estimate, the following procedur will be applied

1. Compute TE_Observed: Calculate TE(SPY -> TLT) and TE(TLT -> SPY) on the original log-return window using the KSG estimator
2. Generate N= 100 shuffled surrogates, Randomly premute the SPY return series within the window essentially destroying its time-series structure while preserving its distribution
3. Compute ETE = TE_Observed - mean(TE_Shuffled). This subtraction removes the bias introduced by finite sample size

4. Statistical Floor: A window's ETE estimate will be considered non-significant (i.e., noise) if TE_observed falls below the 95th percentile of the null distribution of shuffled TE values. Such windows will be flagged but retained in the stability analysis, as their instability is itself an informative signal.

5.3 LGC estimation via VAR

Linear Granger causality will be estimated within each rolling window using a bivariate Vector Autoregression (VAR) model. For each window the F-Statistic of the LGC from SPY TO TLT is defined as:

$$F_{s \to B} = \ln(o^2_{Restricted} / o^2_{Unrestricted})$$

where σ²_Restricted is the residual variance of the restricted model (TLT returns predicted by lagged TLT only) and σ²_Unrestricted is the residual variance of the full model (TLT returns predicted by lagged TLT and lagged SPY). The log-ratio representation follows Geweke (1982) and is the standard continuous measure of Granger causality magnitude.

5.4 Lag Selection

Lag length (k) for both LGC and ETE will be selected using the following principled approaches:

LGC (VAR): Lag length will be selected per window using the Akaike Information Criterion (AIC), with a maximum lag of k_max = 5 trading days. The AIC minimizes the tradeoff between model fit and model complexity by penalizing models for using extra parameters. which is especially important in short rolling windows where overfitting risk is elevated.

ETE (KSG): The embedding dimension (the lag parameter equivalent in TE estimation) will be selected using Mutual Information maximization. Specifically, the embedding lag will be identified via the first local minimum of the average mutual information function (Fraser & Swinney, 1986).

6. Sample Size Justification for Rolling Windows

TO BE RESEARCHED

7. Stability Benchmarking

7.1 Metric A: Coefficient of Variation (Signal Volatility)

Imagine plotting all your 192 rolling-window ETE estimates from 2022. CV tells you how tightly clustered those estimates are around their center. A CV of 0.15 means the typical estimate deviates from the mean by 15% of the mean. A CV of 0.80 means the estimates are scattered wildly and 80% of the mean in typical deviation.

A high CV in 2022 means the method is producing a noisy, inconsistent signal. Each new rolling window gives you a substantially different answer about how much information is flowing between stocks and bonds. You can't trust any single window's estimate because it will likely look very different from the next window's estimate. A low CV means the signal is tracking something real and stable even as the regime shifts.

The Coefficient of Variation is computed using the standard deviation divided by the mean

$$\mathrm{CV} = \frac{\sigma}{\mu}$$

7.2 Metric B: Kurtosis of the Estimate Distribution

Kurtosis measures the shapes of the tail

Calculation of Kurtosis

To calculate kurtosis in statistics, you can follow these steps:

1. Compute the Mean (μ): Calculate the arithmetic mean of the dataset.

2. Compute the Variance (σ^2): Calculate the variance of the dataset, which is the average of the squared differences from the mean.

3. Compute the Standard Deviation (σ): Take the square root of the variance to find the standard deviation.

4. Compute the Fourth Moment (μ_4): Calculate the fourth moment of the dataset, which is the average of the fourth power of the differences from the mean.

5. Compute Kurtosis: The formula for calculating kurtosis is:

$$\text{Kurtosis} = \frac{\mu_4}{\sigma^2}$$

Sometimes, we might also see a version of kurtosis that subtracts 3 from this calculation. This is called excess kurtosis

6. It subtracts 3 because the kurtosis of a normal distribution is 3. So the formula becomes:

$$\text{Excess Kurtosis} = \frac{\mu_4}{\sigma^2} - 3$$

Excess kurtosis greater than zero indicates a leptokurtic (heavy-tailed) distribution, implying that the estimator produces sporadic extreme spikes. A high-kurtosis signal is more likely to generate false causal signals during market stress events. Lower kurtosis in ETE relative to LGC would indicate that ETE is more robust to the outlier shocks that characterize the 2022 environment.

High kurtosis in the LGC estimate series tells you that these spikes are happening far more frequently than a stable model should produce. It tells you the method is hypersensitive to extreme market events. It sees a big shock in the data and overreacts, producing a wildly different causal estimate for that window before snapping back to normal in the next window.

High kurtosis in an estimator's rolling output means it is shock-sensitive. Lower kurtosis means the estimator absorbs the shock without overreacting, producing a more measured and trustworthy signal.

7.3 Metric C: Directional Consistency (DC)

We want to measure how frequently the dominant causal direction remains unchanged between consecutive rolling windows. Thus, we want to measure how frequent the signal flips. A signal that frequently flips could be an unstable detector for a regime shift.

To be Researched

8. Algorithmic Implementation

8.1 Creation of the ETE-RDA and LGC-RDA

The regime detection algorithm detects whether or not the current stock-bond relationship is in a normal or disrupted regime based on the correlation. If the signals given by ETE-RDA and LGC-RDA exceed a statistical threshold, it then signals a regime shift. It reads the daily causal signal and decides whether the market is behaving normally or if it is in a disrupted regime.

The algorithm first establishes a baseline using the 2021 pre-shift period. It then computes the mean and standard deviation of the rolling causal signal for ETE and LGC during the calm reference period. This baseline will represent what a normal information flow looks like before the crash.

A threshold will then be sent at a set standard deviation above the baseline mean. Any signal reading above this threshold will be considered an anomaly relative to the 2021 reference

To prevent the algorithm from reacting to short term noise, we add a persistence filter. The signal must remain above the threshold for at least 10 rolling windows.

When these conditions are met the algorithm will flip a binary flag from 0 to 1, declaring a regime shift has occurred.

8.2 Creation of Dynamic Allocation of Assets based on ETE-RDA and LGC-RDA signals

The Regime Detection Algorithm will be implemented into the Dynamic Asset Allocation Algorithm that changes the weights of assets to minimize drawdown and risk created by volatility.

When the RDA reads 0, the portfolio will maintain the standard 60/40 allocation.

When the RDA reads 1, the portfolio changes the weights of the portfolio allocation. For example a 30/70 allocation.

Three portfolio are constructed in total for comparison: the ETE-RDA driven portfolio, the LGC-RDA driven portfolio, and the static 60/40 benchmark. The static benchmark is the baseline against which both adaptive strategies are evaluated

8.3 Benchmark ETE-RDA, LGC-RDA and static 60/40 algorithm using metrics

We utilize three metrics to determine which strategy best managed risk during the 2022 regime shift.

Volatility measures how much the portfolio's daily returns fluctuated over the test period. A lower number means the strategy produced a smoother and more stable return stream.

Maximum Drawdown measures the worst peak-to-trough loss experienced during the period. It measures how much the portfolio value was destroyed at the worst part of the period.

Sharpe Ratio measures returns earned per unit of risk taken. It justifies whether a strategy's returns justify the volatility it produced. A higher ratio means more efficient risk-adjusted performance

Summary

Phase	Step	Method	Output
Data Prep (2)	**Stock and Bond**	**SPY, TLT Daily Adj**	**Raw price values**

	Data selection	**close**	
Data Prep (2)	**Return Computation**	**Log-returns = ln(P_t / P_{t-1})**	**Stationary return series**
Diagnostics (4)	**Stationarity**	**ADF + KPSS per window**	**VAR in a non-stationary time series produces spurious regressions**
Diagnostics (4)	**Cointegration**	**TBD**	**VECM vs VAR decision in LGC**
Diagnostics (4)	**Normality**	**Rolling Jarque-Bera**	**Time-series of p-values**
Estimation (5)	**LGC Computation**	**VAR + lag selection algorithm**	**LGC signals given k windows and n time lags**
Estimation (5)	**ETE computation**	**KSG + Surrogate testing**	**ETE signals given k windows and n timelags**
Stability (7)	**CV Calculation**	**Standard dev / Mean of rolling estimates**	**CV_ETE vs CV_LGC**
Stability (7)	**Kurtosis Calculation**	**4th moment of estimate**	**Kurt_ETE vs Kurt_LGC**
Stability (7)	**Directional Consistency**	**Flip_count formula**	**Derived DC**
Algorithm Implementation (8)	**Creation of Algorithm**	**TBD**	**ETE-RDA and LGC-RDA**
Algorithm Implementation (8)	**Dynamic Allocation**	**TBD**	**Algorithm that changes the weights of assets based on RDA signals**
Algorithm Benchmarking (8)	**Algorithm Metrics**	**Formulas**	**Maximum drawdown, Portfolio Volatility, and sharpe ratio**

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